

DATA VORTEX 2026

ROUND 4 — TECHNICAL & METHODOLOGY REPORT

Architectural Integration of Data Recovery, Relational Queries, NLP, and Social Signals

September 2026 | Production Build

1. Executive Summary & Objective

Round 4 represents the final synthesis milestone of the DATA VORTEX 2026 competition. The technical objective is to integrate all completed stages—Round 1 data recovery and SQL relational core, Round 2 semantic NLP pipelines, and Round 3 social signal dynamics—into a unified, portable, and interactive analytical product.

2. Multi-Round Data & Asset Architecture

To preserve scientific provenance and eliminate redundancy, Round 4 enforces a strict zero-duplication architecture. All canonical datasets and trained pipelines are dynamically resolved relative to the repository root without machine-specific absolute drive paths.

Milestone	Canonical Relative Path	Verified Volume	Access Protocol
Round 1 P1	Round-1/Phase-1/data/cleaned/Social_Engine_Posts_Cleaned.csv	12,000 rows	Read-Only Pandas
Round 1 P1	Round-1/Phase-1/data/cleaned/Social_Engine_Users_Cleaned.csv	1,500 rows	Read-Only Pandas
Round 1 P2	Round-1/Phase-2/data/data_vortex.db	13,500 rows	SQLite URI (?mode=ro)
Round 2	Round-2/models/sentiment_label_pipeline.pkl	TF-IDF + LogReg	Joblib Pipeline Load
Round 2	Round-2/models/topic_category_pipeline.pkl	TF-IDF + LinearSVC	Joblib Pipeline Load
Round 3	Round-3/data/processed/round3_recommendation_algorithm_reactions.csv	25,415 rows	Read-Only Pandas

3. SQL Relational Core Integration

The production SQLite database at Round-1/Phase-2/data/data_vortex.db is opened strictly using SQLite URI with ?mode=ro. The database integrity passes PRAGMA checks with zero foreign-key violations across 1,500 user rows and 12,000 post rows. The interface directly executes the three authoritative challenge queries: Easy (E2: Top 10 engaged posts), Medium (M1: Location engagement aggregation), and Hard (H2: Creator rank within location using DENSE_RANK window functions).

4. Machine Learning Inference Engine

Round 4 dynamically loads the canonical scikit-learn pipelines trained in Round 2. The primary task is 3-class sentiment classification (Negative, Neutral, Positive) using TF-IDF word unigrams/bigrams and Logistic Regression. True class probabilities are extracted via predict_proba. The secondary task is 4-class topic classification using TF-IDF and Linear Support Vector Classification (LinearSVC). Raw decision margins are provided alongside an explicit softmax approximation, ensuring absolute calibration honesty without fabricating confidence.

5. Signal Processing & Gazetteer Extraction

Social signals from Round 3 track public reactions to recommendation algorithm changes over a 15-year observation window (February 2011 to September 2026). Platform and product mentions are extracted via a documented gazetteer, with strict contextual regex matching for 'X' to avoid single-letter false positives.

6. Portability & Git Integrity

The build incorporates dynamic path resolution from Path(__file__).resolve(). Zero hardcoded machine paths exist. Git history remains immutable, and no git write operations were executed during this milestone.